

Security Level:

Serving Platform

Luca Terracciano luca.terracciano@hauwei.com

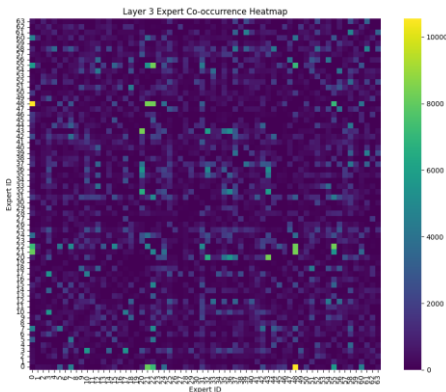
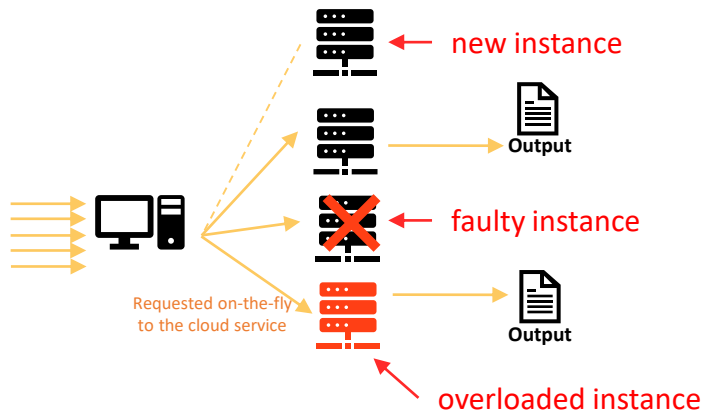
04/06/2026

Rev: A.1

www.huawei.com

Serving Platform: challenges

- We spent last few months working on a federated runtime system targeting LLM inference
- Major challenges & goals:
 - **Scaling & Resource Utilization:** workloads can have major spikes. We want to dynamically maximize the resource utilization + being able to add/remove resources on demand
 - **Latency & Topology-aware placement and scheduling:** the runtime overhead should be negligible (like PTO serving design philosophy states). We want to minimize e2e latency by reconfiguring the deployment based on available resources and application-specific patterns
 - **Reliability & Fault tolerance:** nodes can fail, disrupting the system. We want to handle faults in the system without the need to restart it (zero downtime)



Co-activation of experts in DeepSeekV2

Reconfigurable deployments enable to adjust model/kernels placement to maximize performance

Serving Platform: Architecture

Goal:

- Keep the shortest path to execution (PTO Serving design philosophy)
- Add platform capabilities to make the system more robust and scalable

Core ideas:

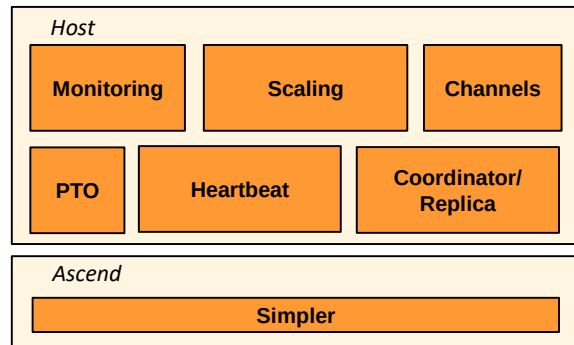
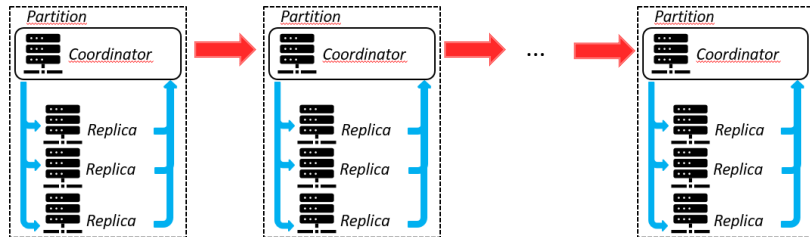
- Serving module split in:
 - **Platform (VNL)**: ramp up the system, establishing communication channels, replication and fault-tolerance mechanisms.
 - **Model (HQ)**: any LLM-specific features. E.g., kv-cache management, request-batching etc.
- Organize the system as a set of partitions, each one responsible for executing a subset of the model. Each partition is executed by Simpler

Single node: install a set of independent modules: **application** + **platform**

- **Simpler/PTO (Ascend runtime)**
- **Heartbeat**
- **Channel management**
- **Monitoring**
- **Coordinator-Replica**
- **Scaling**

Coordinator-Replica deployment: helps manage complexity when scaling the deployment

- Reduces channel
- Fully informed load balancing
- Seamless scaling



Types of parallelism:

- Pipeline parallelism: support out-of-the-box
- Expert parallelism
- Batch parallelism
- Tensor parallelism

We leave to PTO Serving module the others